

Uniform Finite-Field Route Profiles

Pole-Preimage Classification and Characteristic-Aware Stability in Marked Modular Carriers

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Paper XXI of the RIME program. An independently scoped theorem paper on finite marked carriers and routed composition.

Abstract

Problem. Boolean support of each one-step sector block does not determine whether the corresponding labelled routed product is nonzero. For the marked projective carrier over a finite field, the missing information is encoded by the collision pattern of prefix poles.

Approach. For the declared projective action on $X_F = \mathbf{P}^1(F)$, with marked sectors $C_0 = \{\infty\}$ and $C_1 = F$ and labelled alphabet (S, R, R^{-1}) , preimages of ∞ under word prefixes classify each supported route. The same prefix data define fixed-field survivor automata and fixed-depth generic profiles.

Results. Depth two has 45 supported labelled candidates and exactly 14 zero routes for every finite field with at least three elements. Depth three has 216 candidates and

$$|Z_3(F)| = 115 + \mathbf{1}_{\text{char } F=2} + 191\mathbf{1}_{|F|=2} + \mathbf{1}_{|F|=3}.$$

At every positive depth, survivor sets admit an exact prefix-pole classification. It yields the candidate count $3^d F_{d+3}$, fixed-field rational transfer series, and stabilization of the complete labelled depth- d zero-route set whenever $|F| > d$ and $\text{char } F \notin E_d$.

Boundary. Every result is relative to the declared action, labelled alphabet, marked partition, order convention, and route semantics. The fixed-field automaton and reduced rational function may depend on the field. No field-independent automaton, all-depth scalar zero-count formula, presentation invariance, Hecke or modular-form interpretation, RG universality, or spectral zero-mode claim is made.

Keywords: finite fields; projective line; routed products; prefix poles; finite automata; rational generating functions; characteristic stability.

Notation Table

Symbol	Meaning
F, q	finite field and its cardinality
$X_F = \mathbf{P}^1(F)$	marked projective carrier $F \cup \{\infty\}$
C_0, C_1	singleton and finite marked sectors
$A = (S, R, R^{-1})$	ordered labelled projective alphabet
Q_s	projector onto sector C_s
G_k	composition of the first k word letters
$\tilde{G}_k$	declared integral lift of the k th prefix
$v_k(w) = (d_k, -c_k)$	primitive homogeneous vector for the k th prefix pole
$z_k(w) = G_k^{-1}(\infty)$	prefix pole of word w at position k
$\text{Surv}_F(w, s)$	initial states surviving the complete route
$B_d(F), N_d(F), Z_d(F)$	supported, nonzero, and zero-route sets or counts at depth d
Z_d^{gen}	fixed-depth generic zero-route profile
D_d, E_d	prefix determinant spectrum and exceptional characteristics

1 Introduction

This paper studies when a Boolean-supported sector route on a finite projective carrier survives ordered composition. The complete object is

$$(X_F, \rho, A, (C_0, C_1), \prec, \text{Route}),$$

where ρ is the declared permutation action, A retains its labels, $\prec$ fixes composition order, and **Route** fixes the sector-path semantics. None of these components is inferred from the others.

The central theorem classifies a route by the equality and exclusion pattern of its prefix poles. Two consequences follow in different directions:

$$\text{prefix-pole semantics} \Rightarrow \begin{cases} \text{fixed-field survivor automata and rational series,} \\ \text{fixed-depth generic profiles and exceptional characteristics.} \end{cases}$$

The finite-state translation is field-relative and does not imply a uniform automaton. The generic profile is depth-relative and does not imply one all-depth scalar law. Depth-two and depth-three formulas are exact instances of the same semantic construction rather than the source of the arbitrary- depth theorem.

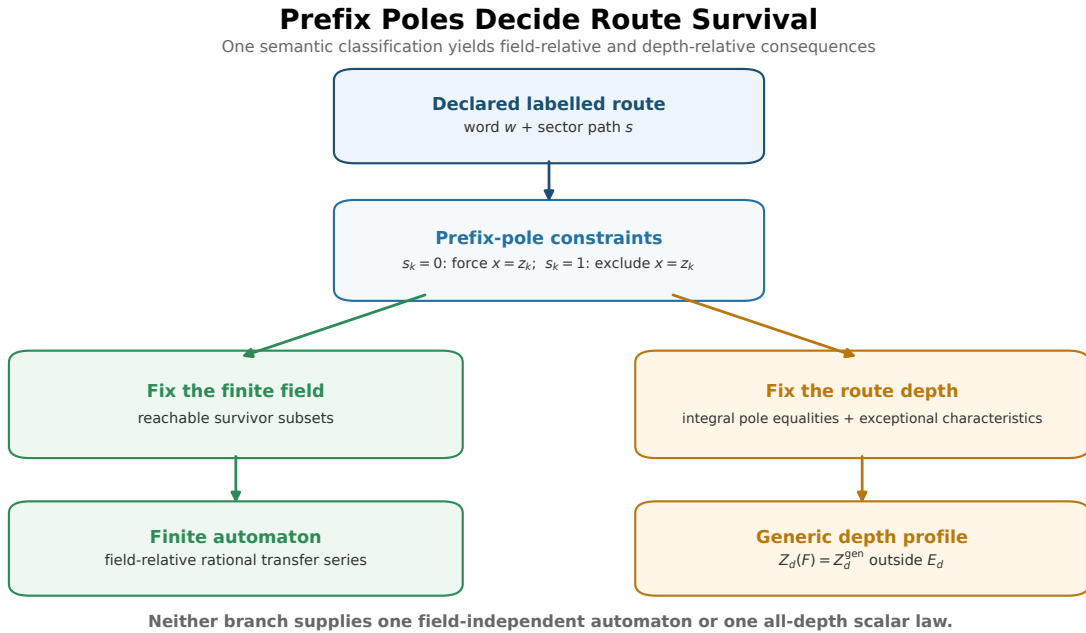


Figure 1. A supported labelled route is reduced to prefix-pole equality and exclusion constraints. Fixing the field yields a finite survivor automaton, while fixing the depth yields a generic profile with exceptional characteristics. Neither branch implies a field-independent automaton or an all-depth scalar law.

The governing distinctions are

$$\begin{aligned} \text{abstract presentation} &\neq \text{permutation action,} \\ \text{permutation action} &\neq \text{marked partition,} \\ \text{Boolean support} &\neq \text{ordered routed product,} \\ \text{zero route} &\neq \text{spectral zero mode,} \\ \text{fixed-depth profile equality} &\neq \text{full semigroup equality.} \end{aligned}$$

Related Work and Novelty Boundary

Finite-field and projective-line arithmetic are standard background [4]. Finite automata, recognizable series, and their rational representations are likewise classical [1]. This paper does not claim those theories. Its contribution is the exact translation from the declared marked modular route semantics to prefix-pole constraints, together with characteristic-aware finite-depth classifications, the induced survivor automata, and source-addressed exact evidence.

Paper XX [2] and this paper belong to the same carrier/route research line, but the results below do not invoke Paper XX's all-depth carrier-accessibility theorem. No nontrivial direct-sum carrier decomposition or Paper XX carrier-preservation certificate is registered for the marked modular family. The classifications instead follow from the explicit projective maps, their poles, and finite-field preimage arithmetic. Thematic succession therefore does not establish theorem dependence.

The same exploratory Schreier source family previously supplied selected finite actions for the exact static realization package of Paper VIII [3]. That promotion certifies a marked finite-permutation SOF realization; it does not classify routed products by prefix poles. The present route-profile evidence is promoted under a separate claim map and receipt closure. Shared source provenance therefore does not transfer theorem ownership or validation scope between the two papers.

A bridge to Paper XX would require a declared $(B, \{V_b\}, \{\Pi_b\})$ and a proof that every labelled generator and marked sector projector preserves every carrier. Without that bridge, this paper is not a corollary, specialization, or application of Paper XX.

2 Marked Modular Carrier

Let F be a finite field and let

$$X_F = \mathbf{P}^1(F) = F \cup \{\infty\}, \quad C_0 = \{\infty\}, \quad C_1 = F.$$

Use the labelled projective maps

$$S(x) = -x^{-1}, \quad R(x) = -(x+1)^{-1}, \quad R^{-1}(x) = -1 - x^{-1},$$

with the fixed integral lifts

$$\tilde{S} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \tilde{R} = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}, \quad \widetilde{R^{-1}} = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix} \in \mathrm{SL}_2(\mathbf{Z}).$$

For the induced prefix lift

$$\tilde{G}_k(w) = \begin{pmatrix} \alpha_k & \beta_k \\ c_k & d_k \end{pmatrix},$$

fix the primitive pole vector

$$v_k(w) = (d_k, -c_k),$$

under the usual projective identification up to sign. Then

$$z_i(w) = z_j(w) \iff \det(v_i(w), v_j(w)) = 0.$$

Unimodularity makes every $v_k(w)$ primitive, so reduction modulo a prime never sends it to the zero vector.

Their values at infinity are

$$S(\infty) = 0, \quad R(\infty) = 0, \quad R^{-1}(\infty) = -1.$$

Their poles are

$$\pi_S = 0, \quad \pi_R = -1, \quad \pi_{R^{-1}} = 0.$$

Labels remain distinct data even when represented maps coincide. A word $[a_1, \dots, a_d]$ acts left to right on states, so its operator product is $P_{a_d} \cdots P_{a_1}$.

Definition 2.1.

For a word $[a_1, \dots, a_d]$ and source-to-target sector path $(s_0, \dots, s_d)$, the route is Boolean-supported when

$$a_k(C_{s_{k-1}}) \cap C_{s_k} \neq \emptyset$$

for every k . It is a zero route when it is supported but

$$Q_{s_d} P_{a_d} \cdots P_{a_1} Q_{s_0} = 0.$$

Equivalently, no carrier state follows the entire declared sector path.

3 Depth-Two Classification

Proposition 3.1.

Every label has the common direct-support matrix, with target rows and source columns,

$$\begin{array}{c|cc} & C_0 & C_1 \\ \hline C_0 & 0 & 1 \\ C_1 & 1 & 1 \end{array}$$

Thus each ordered pair $[a, b]$ has exactly five supported shapes:

$$(1, 0, 1), (0, 1, 0), (1, 1, 0), (0, 1, 1), (1, 1, 1).$$

With nine ordered labelled pairs, the candidate count is $9 \cdot 5 = 45$.

Proof. The displayed matrix is common to all three maps. The supported binary paths of length two are exactly the five listed shapes. $\square$

Theorem 3.2.

For every finite field F with $|F| \geq 3$, the declared carrier has exactly 14 zero routes among 45 supported labelled candidates. Hence

$$\frac{|Z_2(F)|}{|B_2(F)|} = \frac{14}{45}.$$

The shape decomposition is

$$(0, 1, 0) : 4, \quad (1, 1, 0) : 5, \quad (0, 1, 1) : 5.$$

Proof. At infinity the first label gives 0, 0, or -1 for S , R , or R^{-1} . From the intermediate value 0, the second labels (S, R, R^{-1}) give $(\infty, -1, \infty)$; from -1 , they give $(1, \infty, 0)$.

The table supplies four zero routes of shape $(0, 1, 0)$ and five of shape $(1, 1, 0)$, because the actual two-step value lands in the wrong target sector. For shape $(0, 1, 1)$, the five remaining label pairs require, after pole exclusions,

$$S(x) = 0, \quad S(x) = 0, \quad R(x) = 0, \quad R(x) = 0, \quad R^{-1}(x) = -1.$$

These reduce to the impossible finite-field equation $x^{-1} = 0$. The other four source-finite cases have witnesses 1, 0, -1 , and -1 .

For $(1, 0, 1)$, the first-letter pole supplies a witness. For $(1, 1, 1)$, at most two finite inputs are excluded by the two prefix poles. Since $|F| \geq 3$, one finite witness remains. The five shapes exhaust all candidates, giving $4 + 5 + 5 = 14$. $\square$

Corollary 3.3.

For every odd prime p , the modular action on $\mathbf{P}^1(\mathbf{F}_p)$ has exactly 14 zero routes among 45 candidates under this alphabet, partition, and route contract. The prime assumption is stronger than needed: the theorem is finite-field relative and includes extension fields of cardinality at least three.

4 Depth-Three Classification

For $[a, b, c]$, write

$$v_a = a(\infty), \quad \pi_a = a^{-1}(\infty), \quad e_{ab} = b(v_a).$$

The eight supported sector shapes are

$$0101, 0110, 0111, 1010, 1011, 1101, 1110, 1111.$$

Theorem 4.1.

The route is nonzero exactly under the following shape-relative criteria:

0101	$v_a = \pi_b$
0110	$v_a \neq \pi_b$ and $e_{ab} = \pi_c$
0111	$v_a \neq \pi_b$ and $e_{ab} \neq \pi_c$
1010	$v_b = \pi_c$
1011	$v_b \neq \pi_c$
1101	$v_a \neq \pi_b$
1110	$v_b \neq \pi_c$ and $e_{ab} \neq \pi_c$
1111	some finite x avoids all three prefix poles

Proof. The first seven rows trace the unique singleton-sector state. A finite-to-infinity step forces the preceding state to the corresponding pole; a finite-to-finite step excludes that pole. The all-finite row requires $x \neq \pi_a$, $a(x) \neq \pi_b$, and $b(a(x)) \neq \pi_c$.

Its forbidden set is the union of at most three finite prefix preimages. $\square$

Theorem 4.2.

For every finite field F , with $q = |F|$,

$$|Z_3(F)| = 115 + \mathbf{1}_{\text{char } F=2} + 19\mathbf{1}_{q=2} + \mathbf{1}_{q=3}.$$

There are always 216 supported candidates.

Proof. The first seven criteria reduce to equalities among 0, 1, -1 , and infinity. The relevant collapse is $1 = -1$ in characteristic two. For 1111, at most three finite inputs are forbidden. Thus no zero occurs for cardinality at least four in characteristic two or at least five in odd characteristic. In $\mathbf{F}_3$, $[R^{-1}, S, R]$ has three forbidden inputs and gives the unique all-finite zero. In $\mathbf{F}_2$, exactly eight of the 27 words retain a finite witness, yielding 19 all-finite zeros. Summing the shape contributions gives the formula. $\square$

Exact shape decomposition. *The corresponding zero-route histograms are:*

regime	0101	0110	0111	1010	1011	1101	1110	1111	total
$\mathbf{F}_2$	12	22	20	12	15	15	20	19	135
$\mathbf{F}_3$	12	23	19	12	15	15	19	1	116
characteristic 2, $q \geq 4$	12	22	20	12	15	15	20	0	116
odd characteristic, $q \geq 5$	12	23	19	12	15	15	19	0	115

The scalar total is not the complete invariant: $\mathbf{F}_3$ and characteristic-two fields of cardinality at least four both have total 116 but different shape histograms.

5 Arbitrary-Depth Semantic Theory

For a word $w = [a_1, \dots, a_d]$, put

$$G_0 = \text{id}, \quad G_k = a_k \cdots a_1, \quad z_k(w) = G_k^{-1}(\infty).$$

Thus $z_k(w)$ is the initial point whose k th prefix image is infinity. For a sector path $s = (s_0, \dots, s_d)$, define

$$I_0(s) = \{k : s_k = 0\}, \quad I_1(s) = \{k : s_k = 1\},$$

and

$$\text{Surv}_F(w, s) = \{x : G_k(x) \in C_{s_k} \text{ for every } k\}.$$

Theorem 5.1.

For every finite field, every positive depth, every labelled word, and every sector path,

$$\text{Surv}_F(w, s) = \bigcap_{k \in I_0(s)} \{z_k(w)\} \cap \bigcap_{k \in I_1(s)} (X_F \setminus \{z_k(w)\}).$$

Consequently:

1. *If $I_0(s)$ is empty, then*

$$\text{Surv}_F(w, s) = X_F \setminus \{z_0(w), \dots, z_d(w)\}.$$

2. *If $I_0(s)$ is nonempty, the survivor set is a singleton exactly when all forced poles z_k , $k \in I_0(s)$, are equal to one point z and no forbidden pole z_j , $j \in I_1(s)$, equals z . Otherwise it is empty.*

Hence a supported route is zero exactly when the corresponding condition above has empty survivor set.

Proof. For every initial point x , one has $G_k(x) \in C_0$ exactly when $x = z_k(w)$, and $G_k(x) \in C_1$ exactly when $x \neq z_k(w)$.

Intersecting these conditions over all prefix positions gives the displayed identity. The two cases exhaust the possible sector paths. $\square$

Scope. This is an all-depth semantic classification, not an all-depth scalar count. The number and collision pattern of the prefix poles can still vary with the word, depth, and field.

5.1 Generic Depth- d Profile

Evaluate the fixed integral lifts over $\mathbf{Q}$. For a word w , declare prefix indices equivalent when their rational projective poles agree:

$$i \sim_w j \iff \det(v_i(w), v_j(w)) = 0.$$

Let Z_d^{gen} be the supported labelled routes that are zero under this integral prefix-pole equality pattern. This definition is combinatorial: it uses the finite word/sector-path indexing and the rational pole-collision relation, not a selected finite field.

No declared letter fixes infinity. Hence consecutive prefix poles are distinct, so no pole-equivalence class contains consecutive indices. Each class therefore defines a Boolean-supported itinerary whose zero sectors are exactly the indices in that class.

If $c(w)$ is the number of equivalence classes among the $d + 1$ prefix poles, then the generic nonzero itineraries for w are exactly:

1. one itinerary for each prefix-pole class, with zeros at that class; and
2. the all-finite itinerary, represented by every rational point outside the finite prefix-pole set.

Therefore

$$|Z_d^{\text{gen}}| = 3^d F_{d+3} - \sum_{w \in A^d} (c(w) + 1).$$

This is the distinguished scalar sequence for future growth questions. It is not yet supplied with a closed form or asymptotic law.

Corollary 5.2.

Fix a finite field F . A survivor state is a pair (s, A) with $A \subseteq C_s$. Begin in either $(0, C_0)$ or $(1, C_1)$. For a label a and an admitted target sector t , define

$$\tau_{a,t}(s, A) = (t, a(A) \cap C_t),$$

where the common support template admits every sector transition except $0 \rightarrow 0$. Let S_F be the states reachable from the two initial states. Then S_F is finite, with the elementary bound

$$|S_F| \leq 2 + 2^{|F|}.$$

Empty-survivor states are retained rather than pruned. A Boolean-supported route may become semantically dead before its terminal step and must still contribute to $B_d(F)$ and $Z_d(F)$.

Moreover the second component after reading a route prefix is exactly its current-image survivor set. Therefore the route is nonzero exactly when its terminal survivor component is nonempty.

Proof. The transition identity follows directly by applying the next permutation and then imposing the next sector projector. Induction on prefix length proves the survivor assertion. There are two subsets of the singleton sector and $2^{|F|}$ subsets of the finite sector, so the reachable state set is finite with the stated bound. $\square$

Transfer setup. Let M_F be the nonnegative integer transition matrix of this automaton, counting labelled (a, t) transitions. Let u_F select the two initial states, $\mathbf{1}$ be the all-state column, and n_F indicate states with nonempty survivor component.

Corollary 5.3.

For every $d \geq 0$,

$$B_d(F) = u_F M_F^d \mathbf{1}, \quad N_d(F) = u_F M_F^d n_F, \quad Z_d(F) = B_d(F) - N_d(F).$$

In particular,

$$B_F(z) = u_F (I - z M_F)^{-1} \mathbf{1}, \quad N_F(z) = u_F (I - z M_F)^{-1} n_F,$$

and $Z_F(z) = B_F(z) - N_F(z)$

are rational functions in $\mathbf{Q}(z)$ admitting numerator and denominator polynomials in $\mathbf{Z}[z]$. This is fixed-field rationality: S_F , M_F , and the reduced rational functions may change with F .

Proposition 5.4.

For every positive depth d , the supported labelled route count is

$$B_d = 3^d F_{d+3},$$

where $F_1 = F_2 = 1$.

Proof. There are 3^d ordered words. A sector path is supported exactly when it has no consecutive zero sectors. The number of binary strings of length $d + 1$ without consecutive zeros is F_{d+3} . $\square$

Generating function. For $d \geq 0$, the corresponding candidate generating function is

$$B(z) = \frac{2 + 3z}{1 - 3z - 9z^2}.$$

Theorem 5.5.

Use the fixed integral lifts above. For every word of length d , the induced prefix matrix $\tilde{G}_k(w)$ lies in $\mathrm{SL}_2(\mathbf{Z})$, and its primitive pole vector is $v_k(w) = (d_k, -c_k)$. Define the prefix determinant spectrum

$$D_d = \{ |\det(v_i(w), v_j(w))| : w \in A^d, \ 0 \leq i < j \leq d, \ \det(v_i(w), v_j(w)) \neq 0 \}.$$

Define E_d as its prime support:

$$E_d = \{ \ell \text{ prime} : \ell \mid \Delta \text{ for some } \Delta \in D_d \}.$$

Both sets are finite. If a finite field F satisfies

$$|F| > d, \quad \mathrm{char} F \notin E_d,$$

then its complete labelled depth- d zero-route set is the generic profile:

$$Z_d(F) = Z_d^{\mathrm{gen}}.$$

In particular, any two fields satisfying these conditions have identical labelled depth- d zero-route sets.

Proof. Every prefix matrix is unimodular, so its integral pole vector is primitive and remains a nonzero projective vector after reduction in every characteristic. Outside E_d , reduction therefore preserves every equality and inequality among the finitely many prefix-pole vectors: a projective equality holds after reduction exactly when its integral determinant is zero. By Theorem 5.1 this settles every path containing a zero sector. For the all-finite path, at most $d+1$ prefix poles are forbidden, while $|\mathbf{P}^1(F)| = |F| + 1 > d+1$; hence a survivor exists. The same argument applies to every field satisfying the displayed conditions. $\square$

Corollary 5.6.

For every positive depth,

$$D_d \subseteq D_{d+1}, \quad E_d \subseteq E_{d+1}.$$

Proof. Append any declared letter to a length- d word. Its first $d+1$ prefix matrices and pole vectors are unchanged, so every determinant in D_d also occurs at depth $d+1$. Taking prime support gives the second inclusion. $\square$

Exact finite diagnostics. *The exact computational artifact gives the following arithmetic diagnostics:*

depth d	D_d	E_d
1–2	$\{1\}$	empty
3–4	$\{1, 2\}$	$\{2\}$
5–6	$\{1, 2, 3\}$	$\{2, 3\}$
7–8	$\{1, 2, 3, 4, 5\}$	$\{2, 3, 5\}$
9–10	$\{1, 2, 3, 4, 5, 7, 8\}$	$\{2, 3, 5, 7\}$

This explains, rather than merely records, why characteristic five first separates from the generic profile at depth seven: a nonzero prefix determinant of absolute value five first enters D_7 . The table is an exact finite certificate for the displayed depths; Theorem 5.5 is the general mathematical statement. No closed form for D_d or E_d is inferred from the displayed pattern.

Definition 5.7.

The depth- d profile records supported, nonzero, and zero counts; zero histograms by sector path; per-word local counts; an exact zero-route-set digest; and the exact positive-word image layer at depth d . The word-image layer is separate: profile equality does not imply equality of word images or full semigroup closures.

5.2 Computational Certificate 5.8: Sampled Profiles

The exact certificate samples $p = 2, 3, 5, 7, 11, 13$ through depth five:

depth	candidates	$p = 2$	$p = 3$	$p \geq 5$ in sample
1	9	0	0	0
2	45	18	14	14
3	216	135	116	115
4	1053	810	742	732
5	5103	4374	4152	4094

An extended exact probe separates $p = 5$ from $p \geq 7$ at depths seven and eight, consistently with $5 \in E_7$. These tables remain fixed-depth certificates; they do not supply an all-depth scalar formula or a limiting growth law.

6 Computational Evidence

The retained evidence uses exact finite permutations, finite-field arithmetic, integral matrix lifts, and Python integers. No floating-point eigensolver or thresholded zero test enters the recorded classifications.

The depth-three controls cover $\mathbf{F}_2, \mathbf{F}_3, \mathbf{F}_4, \mathbf{F}_5, \mathbf{F}_7, \mathbf{F}_8$, and $\mathbf{F}_9$. The arbitrary-depth replay checks the prefix-pole classification through depth five over $\mathbf{F}_2, \mathbf{F}_3, \mathbf{F}_5$, and $\mathbf{F}_7$, constructs the complete reachable automata for the declared sample fields, verifies transfer counts through depth ten, and computes E_d through depth ten. These finite checks are Computational Certificates for the registered inputs; they do not replace the manuscript proofs of Theorem 5.1, Corollaries 5.2–5.3 and 5.6, or Theorem 5.5.

The compiled Lean surface covers Theorem 3.2, the complete depth-three shape and count package, and Proposition 5.4. It does not cover the arbitrary-depth prefix-pole theorem, fixed-field automata and rationality, the generic profile, or fixed-depth stabilization and determinant-spectrum monotonicity. Lean type checking and exact Python replay are distinct evidence paths; neither is claimed to derive the other.

The source experiments retain their exploratory artifact identities. Promotion forms new source-addressed evidence objects without rewriting those sources in place.

7 Interpretation Boundaries

These are finite marked-carrier theorems, not invariants of the abstract presentation alone:

same presentation $\neq$ same permutation action,

and the same action with a different marked partition need not have the same route profile.

A route zero is a projector-resolved labelled block vanishing. It is not a Hecke eigenvalue, operator-kernel dimension, or modular-form zero. A future Hecke connection would require an explicit double-coset correspondence, averaging semantics, and an equivariance theorem. Cross-field stability is likewise not an RG fixed point; RG language would require a coarse-graining map, a contract state space, and a declared flow.

8 Claim Status and Boundary

The reader-facing evidence levels are:

Surface	Evidence level	Formalization status
Proposition 3.1, Theorem 3.2, and Corollary 3.3	Theorem	compiled Lean source closure
Theorems 4.1–4.2	Theorem	compiled Lean source closure
Theorem 5.1	Theorem	manuscript proof with exact replay; Lean extension open
Corollaries 5.2–5.3	Theorem	manuscript proof with exact construction; Lean extension open
Proposition 5.4	Theorem	compiled Lean source closure
Theorem 5.5 and Corollary 5.6	Theorem	manuscript proof with exact replay; Lean extension open
D_d , E_d , and sampled route-profile tables	Computational Certificate	source-addressed exact replay
unresolved uniform and asymptotic questions	Research Program	open

No Computational Observation is promoted by this paper: every retained table uses exact finite arithmetic. Formalization status is separate from the reader-facing evidence level; a manuscript theorem does not become a Computational Certificate merely because its proof is not yet in Lean.

No claim is made of arbitrary generator or partition invariance, presentation-only universality, a field-independent finite automaton or state bound, one rational function valid for every field, an all-depth closed scalar formula, profile convergence, Hecke or modular-form content, Selberg or Teichmüller content, RG content, or a spectral interpretation of *zero route*.

The remaining questions concern uniformity and minimization, not existence of fixed-field automata, fixed-field rational series, or fixed-depth stabilization.

9 Conclusion

For the declared marked modular carrier, route survival at every positive depth is controlled exactly by prefix-pole equality and exclusion. This gives uniform depth-two and characteristic-aware depth-three classifications, an all-depth semantic criterion, fixed-field survivor automata and rational series, and fixed-depth large-field stabilization whenever $|F| > d$ and $\text{char } F \notin E_d$. The exact finite computations and the compiled Lean subset support distinct parts of that theorem surface without enlarging it.

The result is representation- and marking-relative. It does not promote Boolean support to composition, a fixed-depth profile to a semigroup invariant, or field-relative rationality to one uniform automaton. The next mathematical problem is the arbitrary-depth structure of the generic route language, not another fixed-depth classification.

10 Outlook

The research order is:

1. **Generic route language.** Study the graded family $\mathcal{Z}^{\text{gen}} = \bigsqcup_{d \geq 0} \mathcal{Z}_d^{\text{gen}}$ and its scalar sequence $z_d^{\text{gen}} = |\mathcal{Z}_d^{\text{gen}}|$.
2. **Prefix determinant algebra.** Determine recurrences, closure properties, or growth laws for D_d , E_d , and the pole-class sum $\sum_w c(w)$.
3. **Automaton quotient and minimization.** Replace raw survivor subsets by the coarsest state equivalence that preserves every future route decision. Fixed-field rationality alone does not imply a uniform automaton.

4. **Generic generating function.** Determine whether $G_{\text{gen}}(t) = \sum_{d \geq 0} z_d^{\text{gen}} t^d$ is rational, algebraic, D-finite, or lies outside those classes.
5. **Asymptotic growth.** Only after the preceding structure is available, study the asymptotic quantity

$$\limsup_{d \rightarrow \infty} (z_d^{\text{gen}})^{1/d}$$

and finer asymptotics.

Alphabet/partition comparison and any separate coarse-graining theory remain independently scoped questions.

Appendix A: Computational Artifacts

The computational package is available under `experiments/paper21/` in the [RIME repository](#). Paths in this table are relative to that directory.

Role	Public entry points
arbitrary-depth exact replay	<code>arbitrary_depth_semantic.py</code> ; <code>validation/validate_arbitrary_depth_semantic.py</code>
source-addressed promotion	<code>validation/promote_route_profiles.py</code> ; <code>validation/validate_route_profiles.py</code>
compiled Lean subset	<code>lean/</code> ; <code>validation/validate_lean_formalization.py</code>
claim/evidence alignment	<code>claim-surface-map.json</code>
release closure	<code>release-manifest.json</code> ; <code>validation/validate_release.py</code>
release receipt	<code>results/route_profiles_v1.release-receipt.json</code>

The promotion, arbitrary-depth, and Lean receipts have distinct scopes. The release receipt binds those records, the manuscript, PDF, producers, validators, imported Lean sources, pinned dependency manifest, and hostile tests. A receipt is excluded from its own ordered closure. Verification by the same paper-owned validator is local closure verification, not independent validation and not a proof of validator trust. The experiment README records the exact receipt and closure digests together with the verification entry points.

The Lean receipt binds only the formalized surface listed in Section 8. It does not certify Theorem 5.1, Corollaries 5.2–5.3 and 5.6, or Theorem 5.5. This paper publishes mathematical results and exact evidence; it is not a protocol contract or a claim of independent validation.

References

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